CSE 209 Computer Architecture: What to Study, in Order

Based on 14 past papers (CSE 209 2022-23 and 2023-24, plus CSE 305 2012-2023) and all of your course slides. Study the topics from top to bottom. **Orange = must-do (about 60% of marks)**, **yellow = high**, blue = lower priority.

#	Topic (slides)	Importance	Focus on these (most repeated questions)
1	**Pipelining: data hazards, forwarding, stalls, branch hazards (Ch 4 Part 2)**	VERY HIGH All 14 papers; mandatory Q1 in Section A	Forwarding unit + equations (double data hazard); count cycles/stalls for lw/add/sw; A=B+C; D=A+E code scheduling; hazard detection unit (load-use); 1-bit predictor nested loop; 'perfect predictor still costs 1 cycle'; flush + stall at the same time
2	**Cache memory (Ch 5)**	VERY HIGH All 14 papers	Three small caches (FA / 2-way / DM, LRU) miss count; direct-mapped byte-address trace; tag bits / total cache bits; multilevel cache CPI (4 GHz, 200 ns, L2, L3); write-through vs write-back
3	**Single-cycle datapath & control (Ch 4 Part 1)**	VERY HIGH All 14 papers; mandatory Q5 in Section B	Add bne/j/jr/jal to the datapath; control signal values for add/sw/beq; why PC+4; upper 4 bits of PC in a jump; single-cycle vs pipelined speedup
4	**Virtual memory, page table, TLB (Ch 5)**	HIGH 60 marks in the CSE 209 papers	TLB + page table trace; page table size (single-level, multi-level, inverted); TLB-cache read/write flowchart; which TLB/PT/cache hit-miss combos are possible; context switch
5	**ALU design (ALU Design slides)**	HIGH All of the last 6 papers	Derive Xi, Yi, Zi for a function table; status bits V/Z/S/C; comparator for A>B, A=B, A<B; why overflow = XOR of the last two carries
6	**MIPS assembly & instruction set (Ch 2)**	HIGH Most marks overall, but spread out	Recursive C to MIPS (fib, fun(a,b), sum); loops and ifs; 32-bit constants with lui/ori; machine code encoding; addressing modes; beq out of range / jump reachability; ll/sc; linking, JIT
7	**Arithmetic: multiply/divide hardware + IEEE 754 (Ch 3)**	MEDIUM 13 of 14 papers	First vs refined multiplier with a 4-bit trace; division step by step; bias, smallest/largest/denormalized values; FP addition steps; float/double MIPS code
8	**Performance (Ch 1)**	MEDIUM Easy numerical marks	CPI by instruction class; 'how much must CPI improve'; target clock rate; Amdahl's law; eight great ideas; design principles
9	**GPU/CUDA, multiple issue, cache coherence (GPU slides, Ch 4 Pt 2, Ch 5)**	LOW-MEDIUM Rare, but repeated word for word	PC-GPU connection; kernel/grid/block mapped to SP/SM; dual-issue loop scheduling + IPC + unrolling; snooping protocol
10	**Multicycle control unit, exceptions (Control Unit slides)**	LOW Occasional	EPC and Cause registers; precise exceptions in the pipeline; FSM/microprogram basics

3-week plan. Week 1: topics 1, 2 and 4 (Ch 4 Part 2 + Ch 5). Week 2: topics 3, 5 and 6 (datapath, ALU, MIPS coding; one recursive function every day). Week 3: topics 7-10, then a timed mock using the 2023-24 paper.

Exam pattern. Section A: pipelining, cache, VM/TLB, GPU (Q1 is mandatory and on pipelining). Section B: datapath, ALU, MIPS, arithmetic, performance (Q5 is mandatory and on datapath + ALU).

Skip: I/O, DMA, interrupts, disk/RAID, buses, interconnection networks, multiprocessor C code. They are only in old CSE 305 papers and not in your slides.